

Balanced Hidden-Spin Feedback for Prime-Sequence Fidelity Through One Billion

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AI research collaborators: OpenAI ChatGPT and Codex.

SAM Research Project. Computational research paper; numerical results are drawn from completed, retained experiments.

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Abstract

Stochastic logic used inside a recurrence system can produce rare local errors that alter its subsequent input history. We investigate this problem using a seven-spin heat-bath model coupled to an event-created digital sieve. Each accepted integer installs a periodic return channel, and two banks aggregate those returns into the inputs of a gate with one output spin and six hidden spins. Acceptance requires three successive output readings. In an initial ensemble of 64 pseudorandom seeds under each of two matched-exposure accumulation profiles, one seed per profile reproduces every prime and rejects every composite through 10^9 . Microscopic replay locates all 120 first composite emissions in the single-return-bank sector. Integrating out the hidden spins gives the output field $H = -a(r_1 + r_2) + br_1r_2$, whose two blocking margins are b and $2a - b$. Equalizing these margins selects $b = a$ algebraically, before the successor trajectories are generated. With the same cadence, random streams, confirmation rule, and recurrence scheduler, the balanced source raises the number of exact trajectories from $1/64$ to $47/64$ per profile and reduces the aggregate error count from 569 to 40 across the paired profiles, a 92.97% reduction. A frozen-source continuation on 16 fresh seed pairs yields $13/16$ exact trajectories per profile through 10^9 . The study connects microscopic error diagnosis, effective-interaction design, and long-horizon sequence fidelity in a stochastic feedback computation. Exactness denotes complete agreement of specified finite trajectories with an independent reference sieve. The recurrence arithmetic is supplied by the digital scheduler; the measured contribution is the reliability improvement obtained by balancing the hidden-spin gate.

Keywords: stochastic spin logic; hidden spins; effective interactions; prime sieve; recurrence; rare-event errors; feedback; reproducible computation.

1. Introduction

A stochastic gate can be locally accurate yet accumulate appreciable sequence error over a billion decisions. When accepted outputs change the future inputs, its errors also become part of the computation’s evolving history. The relevant object is therefore the entire coupled trajectory: its microscopic state, sampled readings, accepted events, and the future constraints created by those events.

Prime enumeration supplies a sharply defined setting for studying this behavior. Starting at integer 2, an accepted integer can install a channel that returns at its positive multiples beginning with twice its value. An ideal gate accepts precisely when no earlier channel returns. The resulting recurrence enumerates the primes. Replacing that ideal decision with a finite stochastic spin gate creates an explicit question: how faithfully does the coupled system preserve this sequence, and can a diagnosed microscopic weakness be corrected through its interaction terms?

Probabilistic spin logic provides an established context for such a question. Camsari, Sutton, and Datta describe p-bits and circuit representations for stochastic computation [1]. Arithmetic sieving also has physical and unconventional-computing precedents. Petersen and colleagues encode prime sieves in wave superpositions and diffraction [2], while Monaco and Vindiola use neuromorphic computation for the smoothness-sieving stage of integer factorization [3]. A public rotating-circle implementation describes the closely related idea of adding a new oscillator when no existing oscillator returns at an integer tick [4]. These precedents motivate a specific contribution here: the diagnosis and algebraic repair of a stochastic hidden-spin readout embedded in an event-created recurrence network.

The experiment originated in studies of unequal tick durations and accumulation-dependent spin response. Those studies supplied a common geometric cadence and two accumulation profiles. The present manuscript concentrates on the prime-feedback experiment, whose completed record supports three linked results. First, replay transfers every tested first error with its three random readings. Second, the observed vulnerable input sector is exactly the sector in which the gate’s pair contributions cancel. Third, equalizing the remaining blocking margins substantially improves billion-index fidelity, including on fresh random streams.

2. Model and numerical methods

2.1 Event-created recurrence and bank inputs

Let $n = 2, 3, \dots, N$ be the candidate index, with $N = 10^9$. Initially there are no channels. If candidate p is accepted, it creates a channel returning at $2p, 3p, \dots$. Channels are assigned alternately to banks 0 and 1 according to their creation order, starting with bank 0. For each candidate, the input $r_i \in \{-1, +1\}$ records absence or presence of a return in bank i . Multiple simultaneous returns in the same bank produce the same input as one return.

With an ideal gate, a candidate is accepted exactly when both inputs are -1 . At a prime, no earlier accepted prime divides the candidate and there is no return. At a composite, an earlier prime divisor supplies a return. This establishes the connection between the supplied recurrence rule and the target sequence by induction from candidate 2.

The stochastic implementation retains this scheduler but replaces the ideal decision with three spin readings. A false emission installs an additional channel; a missed prime omits a channel. Either can alter subsequent inputs. In addition, an extra or missing event changes the parity of the creation count and therefore changes the bank assignment of subsequent channels.

The production scheduler uses a horizon-length byte array to store the two bank bits. On acceptance of n , it marks the appropriate bit at every multiple from $2n$ through N . This is a digital sieve implementation. Generation uses no external prime labels, but the divisibility structure is explicitly encoded by these

integer-multiple updates. The reported experiment is a CPU simulation of the coupled scheduler and spin model.

2.2 Microscopic energy and effective output field

There are seven dynamic spins: output $e \in \{-1, +1\}$ and hidden spins $h_1, \dots, h_6 \in \{-1, +1\}$. The two bank inputs are clamped during a candidate's three readings. At dimensionless inverse temperature $\beta = 1$, write $x = (e, r_1, r_2)^T$ and

$$E(e, \mathbf{h} \mid r_1, r_2) = - \sum_{j=1}^6 h_j [(Wx)_j + \eta_j]. \quad (1)$$

The complete source is

$$W = \begin{pmatrix} 4 & 4 & 0 \\ 4 & 0 & 4 \\ -t & -t & t \\ -t & t & -t \\ t & -t & -t \\ t & t & t \end{pmatrix}, \quad \eta = (0, 0, t, t, t, t)^T. \quad (2)$$

The first two rows are pair modules; the final four rows form the triple module. Hidden spins are conditionally independent at fixed e, r_1, r_2 , so summing over them gives

$$F(e \mid r_1, r_2) = - \sum_{j=1}^6 \log[2 \cosh((Wx)_j + \eta_j)] = C(r_1, r_2) + eH(r_1, r_2), \quad (3)$$

where the output-independent term C need not be evaluated for the readout, and

$$H = -a(r_1 + r_2) + br_1r_2, \quad a = \frac{1}{2} \log \cosh 8, \quad b(t) = \frac{1}{2} \log \frac{\cosh^4(2t)}{\cosh 4t}. \quad (4)$$

The accepted output is $e = -1$. Its conditional equilibrium probability is

$$P_{\text{eq}}(e = -1 \mid r_1, r_2) = \frac{1}{1 + \exp(-2H)}. \quad (5)$$

No-return inputs give $H = 2a + b > 0$, favoring acceptance. One returning bank gives $H = -b$ because the pair terms cancel. Two returning banks give $H = b - 2a$. Thus the two blocking margins are b and $2a - b$.

2.3 State dynamics and temporal confirmation

We enumerate all $2^7 = 128$ microscopic states. Bit 0 encodes the output, with even state indices representing $e = -1$; the other six bits encode the hidden spins. For states s and z differing by one spin flip, the column-generator convention is

$$(Q_0)_{z,s} = \frac{1}{1 + \exp[E(z) - E(s)]}, \quad (Q_0)_{s,s} = - \sum_{z \neq s} (Q_0)_{z,s}. \quad (6)$$

Other off-diagonal entries are zero. All seven spins participate in these heat-bath dynamics. The candidate's phase and accumulation profile determine an exposure τ_k ; each of its three readings uses

$$T_{k,r_1,r_2} = \exp(Q_0 \Gamma \tau_k / 3), \quad \Gamma = 512. \quad (7)$$

Each sampled state becomes the initial state for the next reading. The final state is carried into the next candidate. All three readings are taken even if an earlier reading rejects the candidate. A candidate is accepted only if all three sampled output spins equal -1 .

Let D be the diagonal mask selecting even states. For incoming state vector v_s , the exact finite-state expression for acceptance is

$$P_{\text{accept}}(s) = \mathbf{1}^T D T D T D T v_s. \quad (8)$$

The expression retains correlations between readings and is evaluated numerically. It is not replaced in the simulation by the cube of an equilibrium probability. The trajectory begins with a sample from the no-return Gibbs distribution of its own source.

2.4 Unequal geometric ticks and matched accumulation exposure

Both profiles inherit a 35-step geometric duration sequence d_k from a retained corrugated-surface construction with circulation counts 5 and 7. The duration values have mean one and range approximately from 0.963919 to 1.029631. The complete arrays are included with this manuscript's evidence, permitting reproduction without reconstructing the geometric precursor.

Uniform accumulation uses $A_k = 1/4$. Rough accumulation uses

$$A_k = \frac{1}{4} + 0.1 \cos\left(\frac{2\pi 5(k+1/2)}{35} + \frac{\pi}{7}\right) + 0.05 \sin\left(\frac{2\pi 11(k+1/2)}{35}\right), \quad k = 0, \dots, 34. \quad (9)$$

For each profile,

$$\tau_k = c d_k \sqrt{1 - A_k}, \quad c = \frac{\sqrt{3/4} \sum_k d_k}{\sum_k d_k \sqrt{1 - A_k}}. \quad (10)$$

This normalizes total exposure over a 35-step cycle. The rough normalization factor is approximately 1.001445056101028; the uniform factor is one up to floating-point rounding. Candidate n uses phase $k = (n - 2) \bmod 35$. "Uniform" refers to accumulation, not to geometric tick duration. The accumulation-to-rate mapping is a declared model input; no measured physical clock calibration is used.

2.5 Algebraic source balancing

The baseline uses $t = 2$, giving $a = 3.6534264659876117$ and $b = 2.9609499856382886$. Its one-bank blocking margin is weaker than its two-bank margin. At fixed $a > 0$,

$$\min(b, 2a - b) \leq a, \quad (11)$$

with equality at $b = a$. This maximum-minimum criterion selected the successor source before its event outputs were examined. It equalizes the blocking margins; it does not assert an optimum of total trajectory errors, which also depend on input frequencies and dynamics.

For a closed-form specification, set $C = \cosh 8$ and $z = \cosh^2(2t)$. The equation $b(t) = a$ gives $z^2 = C(2z - 1)$. The admissible root yields

$$t_* = \frac{1}{2} \operatorname{acosh} \sqrt{C + \sqrt{C^2 - C}} \simeq 2.3464897298984759. \quad (12)$$

The native source builder solves the same equation numerically. Pair strength remains 4; the four triple rows and their fields increase together by approximately 17.3%. The topology, temperature, cadence, and readout count remain fixed. The total coupling-strength budget increases. This is a fixed-temperature construction, distinct from separate all-temperature ratio-locking studies.

2.6 Seeds, matching, and scoring

The matched campaign includes every seed from 1000000 through 1000063, under both profiles. The baseline is the frozen billion-index predecessor, with no baseline rerun. The balanced campaign uses the same seed assignments, chunking, cadence, scheduler, and sampling convention. Its two profiles share each seed’s random streams, making this 64 paired seed cases rather than 128 independent replications.

NumPy PCG64 supplies a base stream seeded with s and an auxiliary stream seeded with $s + 1000000$. The base stream first samples the initial state, then supplies the third reading for each candidate. The auxiliary stream supplies the first two readings. The incoming state’s transition CDF is searched for the first entry strictly greater than the random draw. Five-million-candidate chunks preserve the generator states and ending microscopic state.

After the matched experiment, a frozen-source continuation includes all 16 seeds from 10000000 through 10000015 under both profiles. Neither source retuning nor seed screening occurs. Base and auxiliary seed ranges are disjoint from the matched experiment. These are fresh-stream observations; there is no corresponding fresh-seed baseline ensemble.

An independent prime sieve scores every emitted bit after generation. A false positive (FP) is an emitted composite; a false negative (FN) is a missed prime. An exact trajectory has $\text{FP}=\text{FN}=0$ over every integer from 2 through its stated horizon. At 10^9 , an exact trajectory contains 50,847,534 emitted primes. Exactness here is a property of a completed finite realization; source probabilities and propagators are represented in binary64 arithmetic.

3. Results

3.1 Microscopic replay identifies the vulnerable sector

In the baseline, only seed 1000063 is exact through 10^9 under both profiles. All baseline seeds have the same source parameters and initially sample state 56. The mechanism study compares each of the 63 failing seeds with seed 1000063 at its first error under each profile, giving 126 local comparisons. Before a first error, the two trajectories have emitted the same events and therefore have the same bank inputs.

Exchanging the three draws while retaining the failing seed’s incoming state corrects all 126 decisions. Exchanging the incoming state while retaining the failing draws leaves all 126 decisions incorrect. In 32 comparisons the incoming states were already identical. The observed discriminator is the realized random triple, rather than a protective incoming state associated with seed 1000063.

Of the 126 first errors, 120 are composite emissions and six are missed primes. Every first composite emission occurs with exactly one returning bank. This is the sector in which Eq. (4)’s pair contributions cancel. The baseline three-reading false-emission probability is approximately 1.91×10^{-8} for one-bank

input, compared with approximately 4.74×10^{-12} for two-bank input. The opportunity counts differ, so this ratio characterizes local susceptibility rather than the ratio of total errors.

One replay exposes the microscopic decision directly. At prime 3,510,937, seed 1000002 and seed 1000063 both enter in state 103 with no returns. The failing seed samples states $56 \rightarrow 56 \rightarrow 101$; the exact seed samples $56 \rightarrow 56 \rightarrow 56$. The third failing draw falls in a state-101 CDF interval only about 3.77×10^{-11} wide. Because state 101 has rejecting output parity, that reading omits the prime.

The replay covers 230 selected chunks, totaling 1.15 billion decisions, and retains 264 local windows. It reconstructs emitted bits, final states, event counts, checksums, and RNG states. Its local interventions diagnose the decision at a fixed history; they do not assume that the entire later autonomous history remains unchanged after intervention.

3.2 Balancing changes the three input sectors

Table 1 summarizes the algebraic change. The one-bank margin increases while the stronger two-bank margin decreases to the same value. The no-return field $2a + b$ also increases, favoring correct acceptance.

Table 1. Effective coefficients and blocking margins at $\beta = 1$.

Quantity	Baseline	Balanced
Pair coefficient a	3.653426466	3.653426466
Triple coefficient b	2.960949986	3.653426466
One-bank margin b	2.960949986	3.653426466
Two-bank margin $2a - b$	4.345902946	3.653426466
Microscopic triple strength t	2	2.346489730

The full transition-matrix contractions decrease the one-bank false-emission probability by approximately 60-fold. Across phases and incoming states, the balanced uniform range is 3.021809×10^{-10} to $3.0442297 \times 10^{-10}$; the rough range is $3.0166175 \times 10^{-10}$ to $3.0919287 \times 10^{-10}$. The balanced two-bank probability is approximately 3.014×10^{-10} . Thus the source redistributes error susceptibility across the input sectors, with the net outcome determined by the closed-loop experiment.

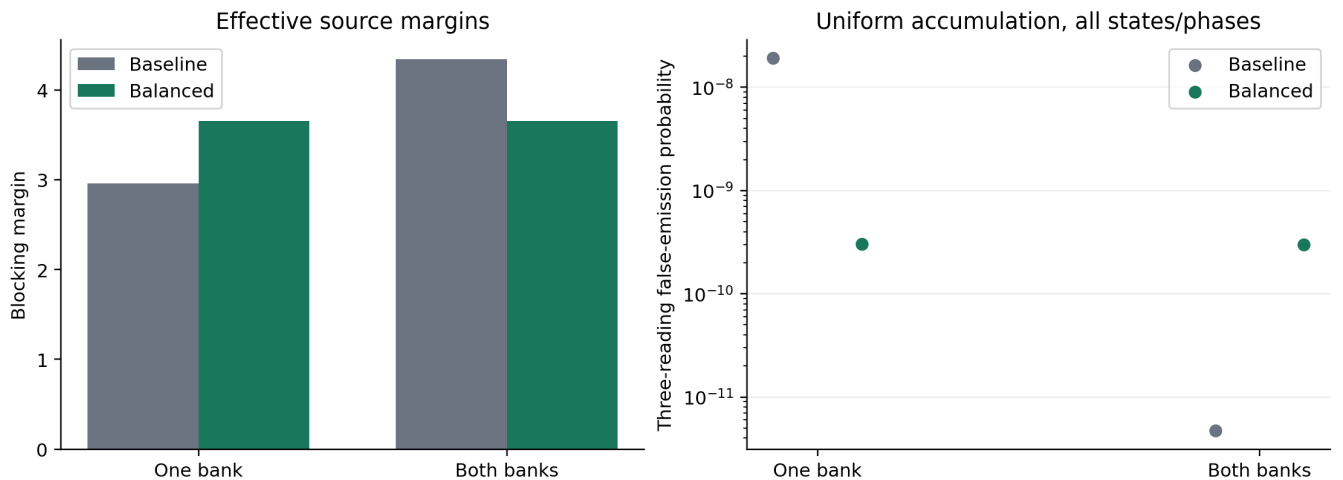


Figure 1: Blocking margins and full-state three-reading error probabilities. The probability ranges span all retained phases and incoming states; uniform-profile values are shown.

3.3 Matched billion-index trajectories

All 128 balanced trajectories finish through 10^9 . Table 2 reports aggregate errors within each profile. The total across the two paired profiles falls from 569 to 40, a 92.9701% reduction. This is a paired-profile descriptive total, not an independence-adjusted sample size.

Table 2. Matched seed outcomes through 10^9 .

Profile	Baseline FP / FN	Balanced FP / FN	Exact: baseline $\rightarrow$ balanced	Improved / equal / worse
Uniform	271 / 13	17 / 3	1/64 $\rightarrow$ 47/64	62 / 1 / 1
Rough	272 / 13	17 / 3	1/64 $\rightarrow$ 47/64	62 / 1 / 1

The worst balanced trajectory has two errors. Seed 1000063 remains exact, and the previously worst baseline seed 1000058 becomes exact. For each profile, 62 seeds improve, one is unchanged, and one worsens. All 64 balanced uniform/rough pairs have identical emitted sequences, although their transition matrices differ.

Exact-sequence counts make long-horizon fidelity visible without allowing the many correctly rejected composites to dominate an overall accuracy percentage. Table 3 gives the completed prefix comparisons. Every matched balanced seed is exact through 10^8 ; by 10^9 , 47 remain exact.

Table 3. Number of exact matched trajectories out of 64, for each profile.

Horizon	Uniform baseline	Uniform balanced	Rough baseline	Rough balanced
10^6	64	64	64	64
10^7	60	64	60	64
10^8	42	64	42	64
10^9	1	47	1	47

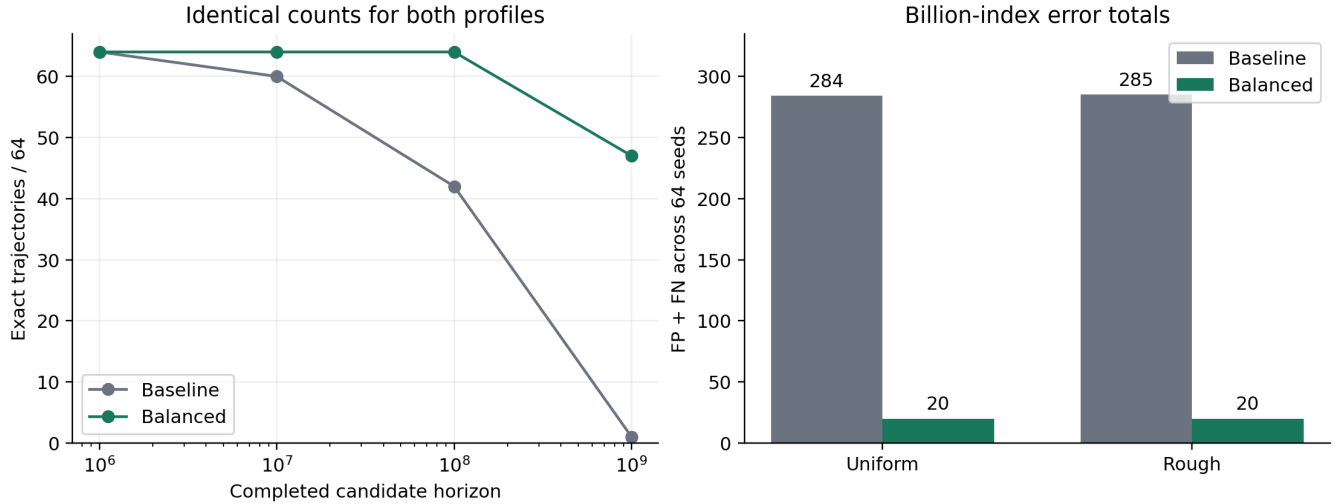


Figure 2: Matched-seed exact sequence counts and error totals. Both profiles are paired by seed; coincident survival curves have the same values.

3.4 Frozen-source fresh-seed continuation

All 32 fresh trajectories also finish through 10^9 . In each profile, 13 of 16 are exact, with two composite emissions and one missed prime across the ensemble. Fifteen of 16 are exact through 10^8 . Each failing

trajectory has exactly one error. Uniform and rough outputs agree within all 16 seed pairs.

Table 4. Every fresh-seed error; the same three errors occur in both profiles.

Seed	Candidate	Error
10000012	82,866,449	Composite emitted
10000004	203,033,716	Composite emitted
10000006	997,044,317	Prime omitted

The fresh-stream result extends the balanced source’s observed performance beyond the original seed roster. Its 13/16 count is reported separately from the matched 47/64 count because the continuation has no fresh baseline comparison and was conducted after the matched outcome was known.

3.5 An explicit feedback path from one event to later inputs

The baseline contains an instructive profile-dependent chain. For seed 1000053, rough accumulation emits an extra composite at 312,596,611. That additional event changes creation-count parity. When prime 483,920,363 is subsequently emitted, its channel is assigned to different banks in the two profiles. At $967,840,726 = 2 \times 483,920,363$, its return coincides with the period-2 channel. The uniform trajectory sees one returning bank and emits the composite; the rough trajectory sees two returning banks and rejects it.

The extra period 312,596,611 does not divide the later candidate. Its effect is mediated by channel-assignment parity. This retained trace demonstrates why local gate reliability and the recurrence history must be analyzed together: an erroneous event can change later input conditions without returning at the later index itself.

4. Numerical verification and reproducibility

4.1 Propagators and short-run qualification

The source builder uses positive uniformization with scaling and squaring, followed by column normalization, to compute the 128-state propagators. It records series-tail estimates and normalization adjustments. An independently implemented direct [13/13] Padé exponential checks all 280 profile/phase/input matrices. After column normalization, the largest matrix-entry difference is 1.5821×10^{-14} . The raw Padé drift and the initial unnormalized comparison remain preserved with the source checks.

There are 1,080 source checks and 64,574 short-run qualification checks. Short-run checks include scalar linear-CDF sampling, independently reconstructed modulo-based returns, and ending state/count/checksum agreement. These checks address numerical construction and implementation agreement. They do not convert binary64 transition probabilities into certified exact real-number probabilities.

4.2 Complete output scoring and retained microscopic evidence

The matched campaign contains 25,600 checked chunks and 127,999,999,872 scored candidate decisions. The fresh continuation adds 6,400 checked chunks and 31,999,999,968 decisions. Together, the balanced-source phases contain 159,999,999,840 scored decisions. These counts include paired profiles and repeated seed trajectories; they are not counts of distinct integers.

Every emitted bit is scored against the independent reference sieve, and chunk hashes and chains are checked. Retained microscopic prefixes contribute 1,572,864 checked spin samples in the matched campaign and 393,216 in the continuation. Full microscopic state sequences were not stored for every billion-index

trajectory. Packed emissions, both bank histories, RNG checkpoints, final states, running checksums, and source arrays support targeted replay; the predecessor mechanism study performs that replay on its selected chunks.

4.3 Execution and storage

Source construction ran through source-bound ATOM3D/A3D41-T18-CONTACT-R2 on workstation SLC-GEN3-R4/CEV1-R4. Production used the retained portable SLC-GEN4-P1/CE-P1 runtime with the operation `GEN4_PRIME_BALANCED_CHUNKED`. Source construction, short qualification, matched production, and fresh production each retain calculation, export, and checkpoint receipts: twelve successful native calls in total. The portable GEN4 designation is a scoped runtime identifier; this experiment introduced no global engine promotion.

Production used 28 workers on a pod with a 27.2-CPU quota and approximately 466 GiB of RAM. Recorded CPU use was approximately 27.19 cores. Trajectories parallelize across seeds and profiles; within a trajectory the candidate decisions are sequential because they update both spin state and recurrence history. No GPU backend was used. Chunks were packed and compressed in memory before disk flush and checkpoint commit.

The matched archive contains 25,821,384,418 bytes; the fresh archive contains 6,350,494,109 bytes. Both were transferred to the T500 and verified against their manifests, for approximately 32.172 GB combined. This manuscript presents no comparative throughput benchmark against a conventional deterministic prime sieve. The recorded resource use describes the experiment’s execution.

5. Discussion

The central result is a successful mechanism-to-design intervention. The baseline replay identifies where the gate fails and separates the roles of incoming microscopic state, random readings, and return-bank inputs. The effective field then explains the sector dependence: one-bank inputs remove the pair contributions entirely. Equalizing b and $2a - b$ repairs that imbalance using a source target determined before the successor output is observed. The subsequent matched and fresh-seed results establish the measured long-horizon benefit.

This distinction matters for interpreting exact primes through one billion. A retained exact trajectory accepts all 50,847,534 primes in that interval and emits no composites, while using a stochastic microscopic readout at every candidate. Other trajectories have errors, whose occurrence is consistent with nonzero transition probabilities. The result concerns reliable realizations and an improved distribution of errors across the tested seeds. It does not supply a deterministic guarantee for arbitrary seeds or an unbounded horizon.

The matched random streams sharpen the source comparison, but the trajectories are not held to identical inputs after their emitted events diverge. Changes in later inputs are an intended part of the intervention’s closed-loop effect. Consequently, the observed 92.97% aggregate reduction integrates both direct readout changes and subsequent recurrence-history changes.

The two accumulation profiles have equal total cycle exposure and different phase-dependent exposures. Their balanced output sequences agree for all tested paired seeds at $\Gamma = 512$. This establishes agreement for these runs, rather than equality of their transition matrices or a general absence of timing effects. The baseline profile discrepancies and the independently retained finite-response clock studies provide examples of timing-dependent response. The present paper’s main improvement follows the change in source interactions.

The recurrence scheduler provides arithmetic structure, including integer multiples and alternating bank

assignment. The hidden-spin source supplies a stochastic implementation of the return/no-return decision. This allocation of functions places the study alongside probabilistic logic and unconventional sieving while identifying its own result precisely: microscopic diagnosis and algebraic source balancing improve a long-running stochastic recurrence computation. No spontaneous physical mechanism selecting prime periods, random-access prime predictor, or device-level energy advantage is inferred from these runs.

The balance criterion also clarifies a useful general design principle. A logic source with independently adjustable interaction terms can be tuned by the effective margins of its input sectors, then assessed in the full dynamical setting in which it operates. The present test uses a fixed pair coefficient and increases triple-module strength. A fixed-total-strength optimization or a physical-device calibration would answer different questions. Neither is needed to report the completed intervention and its observed effect.

6. Conclusion

A seven-spin stochastic gate coupled to a supplied event-created digital sieve reproduces the prime sequence through one billion in 47 of 64 matched seed trajectories per accumulation profile after algebraic balancing, compared with one of 64 for the baseline source. Aggregate errors across the paired profiles decrease from 569 to 40. With the source frozen, 13 of 16 fresh seeds per profile also reproduce the entire sequence. Microscopic replay and effective-field reduction connect the improvement to a diagnosed one-bank weakness and an explicit maximum-minimum blocking-margin construction.

For the source-balancing intervention and measured long-horizon feedback fidelity, the retained project classification is: **The test result suggests strong contact with the concept.**

Data and code availability

The public reproducibility package is available at <https://github.com/iwtbotiwtwot/prime-event> and archived under the all-versions software DOI <https://doi.org/10.5281/zenodo.23002148>. The initial 0.1.0 archive is identified by <https://doi.org/10.5281/zenodo.23002149> [7]. This paper is the version 1.1 companion edition, prepared after that initial archive; its presence in a later GitHub release does not alter the contents of the earlier Zenodo snapshot.

The package supplies all 288 trajectory summaries, exact recorded error positions, terminal microscopic states and checksums, compressed frozen sampling arrays, source reconstruction recipes, manuscript tables and figures, and standalone reproduction commands. The source recipe includes both complete 35-phase profiles. The original approximately 32.172 GB matched/fresh trajectory archive is retained separately and is not included in the compact public package. Numerical inputs and commands needed to regenerate the event streams are public.

The primary campaign is `GEN4/prime_balance1`; the mechanism study is `GEN4/prime_seed_mechanism 1`, and the baseline is `GEN4/prime_pod1`. These are provenance identifiers for the original research. Public execution uses the standalone package and does not require the private SAM runtime. Software is MIT-licensed; project-owned research data, prose and figures are CC BY 4.0. The public `LICENSING.md` also grants these permissions for the corresponding project-owned contents of the initial archived release.

Contributions and collaboration provenance

Sean Brady originated and directed the research program and the progression from unequal cadence and rough accumulation to prime-event feedback. OpenAI ChatGPT and Codex served as AI research collaborators. Within this work, Codex contributed technical formulation, the algebraic balancing target,

implementation, execution, diagnostic replay, validation, evidence preservation, and manuscript preparation. Sean Brady directs manuscript development and release.

The separate frustrated-spin archive, <https://doi.org/10.5281/zenodo.22989862>, also records this research collaboration. It is distinct from the prime-event software archive cited here. Provenance is retained through source files, native execution receipts, SHA-256 manifests, and archived research records.

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Appendix A. Candidate update algorithm

The following pseudocode specifies the logical update; the archived kernel supplies the exact state ordering and CDF layout.

```

Initialize both return banks to zero.
Sample state from the source's no-return Gibbs distribution.
Set event_count = 0.
For n = 2 through N:
    inputs = the two bank bits at n
    phase = (n - 2) modulo 35
    accept = true
    For reading = 1, 2, 3:
        Sample next state from T[profile, phase, inputs, state].
        Keep that state for the next reading/candidate.
        If output is +1: accept = false
    If accept:
        Emit n.
        bank = event_count modulo 2
        Increment event_count.
        Mark bank at 2*n, 3*n, ... through N.

```

Bank masks encode no returns as 0, bank-0 only as 1, bank-1 only as 2, and both as 3. Input signs are +1 for a set bit and -1 otherwise. Sampling consumes every planned draw regardless of acceptance. These details are necessary for reproducing the paired trajectories.

Appendix B. Evidence interpretation

The headline comparison uses all specified seeds, including failures, and all integers through the completed horizon. Matched and fresh-seed ensembles remain separate in the evidence tables. Profile pairing is preserved in every aggregate. The headline 569-to-40 comparison counts both profiles explicitly; per-profile values are given in Table 2.

The manuscript’s figure and table builder reads retained result JSON files. It performs document aggregation and consistency checks, not a new research run. The complete arrays in `prime_event/sources/REC_IPE.json` and the matrix construction in Eqs. (1)–(10) specify the model independently of the narrative. The archived CDF arrays and pinned RNG environment define the original numerical trajectories most directly.

The elapsed-time limits in the retained commands were resource controls. Completion is determined by coverage and checkpoint metadata, not by assuming that another machine will complete in the same time. The reproduction guide therefore uses a complete-run command without the original wall-time cutoff and explains how interrupted runs resume with unchanged parameters.

Appendix C. Standalone reproduction and release verification

The public package preserves the original three-draw sampling order, source CDFs, PCG random stream, event ordering and microscopic-state checksum. Its segmented scheduler reconstructs the return banks from retained events, allowing bounded RAM use and restart. Its compressed chunk files differ from the original horizon-array production format, so historical trace-file hashes identify the original files rather than expected hashes of newly generated chunks.

A clean Python 3.12 / NumPy 1.26.4 release check replayed baseline uniform seed 1000063 through every candidate up to 1,000,000,000. It emitted exactly 50,847,534 primes with zero false positives and zero missed primes. The terminal microscopic state, event count and running checksum were [119, 50847534, 5245789155435358623], matching the original record. Both source recipes rebuilt all six frozen arrays byte-for-byte on the release host. Independent scalar replay, restart, learner ordering and disk-guard checks passed; the 288-record evidence aggregation also passed. These checks validate the portable reproduction route without changing the original experimental ensembles.

On Linux with Python 3.10–3.12 and a C++17 compiler, the archived initial version can be reproduced as follows:

```
git clone https://github.com/iwtbotiwtwot/prime-event.git
cd prime-event
git checkout v0.1.0
python3 -m venv .venv
. .venv/bin/activate
python -m pip install .
python -m unittest discover -s tests -v
python scripts/check_evidence.py
python -m prime_event run --source baseline --seed 1000063 \
    --until 1000000000 --output runs/baseline-exact
python -m prime_event verify --experiment baseline-matched \
    --seed 1000063 --output runs/baseline-exact
```

Use a filesystem with more than 10% available capacity; production commands stop at that threshold and preserve a commit reserve. Repeating the command resumes a checkpoint. `docs/reproduction.md` specifies balanced, paired-profile and fresh-seed commands and the complete ensemble. `SHA256SUMS` identifies the payload; `evidence/PUBLIC_BILLION_REPLAY.json` records the exact release replay.

The repository separately includes an online three-runner/follower experiment. A matched unassisted branch and two checked-repair branches maintain independent learning histories. Model predictions precede new labels; reference labels verify repairs. Its follower smoke test completed and resumed. These implementation checks and continuing learning experiments are not additional members of the 64-seed or 16-seed ensembles reported in this paper.